

EVALUATING PMC STRATEGIES ACROSS PERENNIAL AND SUB-PERENNIAL MALARIA SETTINGS USING MATHEMATICAL MODELLING

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August 12th, 2026

This work was conducted during the presenters' academic affiliations and independently of their current employment.



Malaria Chemoprevention Alliance Research Subgroup Meeting

Wednesday 12th August 2026

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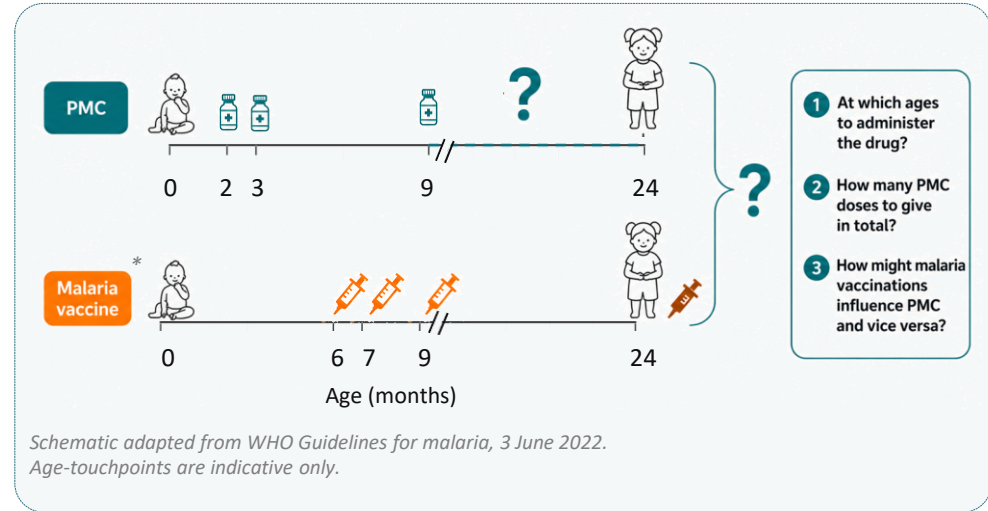


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**Topic: Evaluating PMC Strategies Across Perennial
and Sub-Perennial Malaria Settings Using
Mathematical Modelling**

Malaria prevention in children: protection needs to match age and transmission pattern under consideration of feasible coverages

- WHO estimated 282 million malaria cases and 610,000 deaths globally in 2024, with the **highest burden in young children** <5years of age
- Young children require **continued protection** across **changing age-specific risk** and seasonal transmission patterns.
- **Protection gaps persist:** some children lack access to prevention and treatment, while others remain at high risk despite access.
- **PMC and malaria vaccines can both leverage immunization platforms.** Increasingly flexible delivery options raise key questions around **timing, co-delivery, incremental impact, and prioritization** – and operational effectiveness.

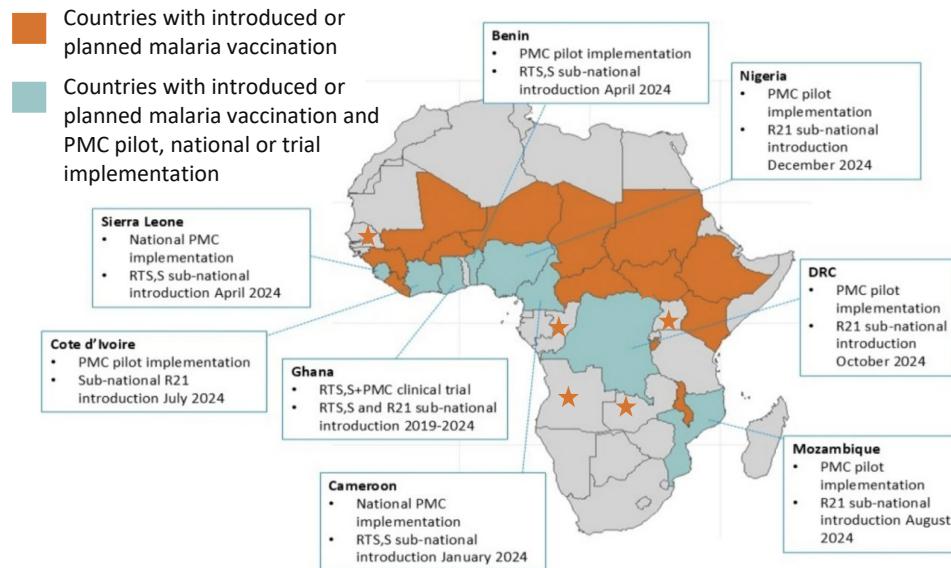


*) Other age-schedules, and delivery modes, such as season-based or hybrid delivery approaches exist

As malaria vaccines scale up, PMC and vaccine delivery need to be coordinated

PMC and malaria vaccination implementation

- **Routine age-based delivery:** Uses existing immunization contacts and systems, but may not align protection with periods of highest transmission.
- **Hybrid age-seasonal delivery:** Combines age-based and seasonal delivery to better match periods of highest risk, at the cost of greater operational complexity.
- **PMC + vaccine co-delivery:** May provide complementary protection across transmission periods, but direct evidence remains limited. Optimal timing, sequencing, and delivery remain key open questions.



Adapted from: **Grant and PMC community of practice (2025)**. A roadmap of priority evidence gaps for the co-implementation of malaria vaccines and perennial malaria chemoprevention.

★ Additional countries as per: International Vaccine Access Center (IVAC), Johns Hopkins Bloomberg School of Public Health. VIEW-hub. www.view-hub.org. Accessed: 1.8.2026.

Which delivery strategy and combinations performs best under different transmission patterns?

Research questions

1. **What is the public health impact of current and proposed age-expanded PMC?**
 - Does extending PMC from 24 to 36 months increase impact while maintaining positive net benefit, under varying coverage and drug-resistance scenarios?
 - How do transmission, drug resistance, coverage, and perennial pattern affect impact?

2. **How does PMC compare with, and complement, malaria vaccination?**
 - How does the maximum achievable impact of standard PMC, age-expanded PMC, and RTS,S compare?
 - What additional impact is gained by combining PMC + RTS,S relative to either intervention alone?
 - What is the expected operational and public health impact of these strategies in a high-burden, perennial-transmission setting?

METHODS AND RESULTS WERE BASED ON THE FOLLOWING PUBLICATION. TO VIEW THE PRESENTED SLIDES, CONTACT THE PRESENTERS.

Sen, S., Braunack-Mayer, L., Kelly, S.L. *et al.* Public health impact of current and proposed age-expanded perennial malaria chemoprevention: a modelling study. *Sci Rep* **15**, 10488 (2025). <https://doi.org/10.1038/s41598-025-93623-z>

Sen S, Schellenberg D, Penny MA. Rethinking a hybrid malaria chemoprevention delivery strategy for children in sub-perennial settings: a modelling study integrating age- and seasonally-targeted delivery. *Malar J*. 2025 Nov 21;24(1):419. doi: 10.1186/s12936-025-05630-0. PMID: 41272819; PMCID: PMC12639754.

Runge M, Stahlfeld A, Ambrose M, Toh KB, Rahman S, Omoniwa OF, Bever CA, Oresanya O, Uhomoibhi P, Galatas B, Tibenderana JK, Gerardin J. Perennial malaria chemoprevention with and without malaria vaccination to reduce malaria burden in young children: a modelling analysis. *Malar J*. 2023 Apr 24;22(1):133. doi: 10.1186/s12936-023-04564-9. PMID: 37095480; PMCID: PMC10124689.

Remaining Questions and Evidence Needs

- Where would PMC + malaria vaccine co-delivery provide the greatest incremental benefit?
- How should PMC schedules be adapted as malaria vaccines scale up?
- What needs to be evaluated specifically for R21/Matrix-M, given that RTS,S was modelled here?
- How should countries balance additional PMC touchpoints, higher coverage, hybrid delivery, and vaccine co-administration?
- What operational data are most urgently needed: dose-by-dose coverage, correlated uptake, missed doses, costs, acceptability, or supply constraints?
- How to harmonize quantitative definition of malaria seasonality to guide chemoprevention strategy deployment?
- Which factors will be important for maintaining health economic benefit for alternative PMC drugs to as SP-resistance increases?